FROB C5 finite-field factor-kernel audit

Validation audit, 22 October 2026

Scope. Frozen `EXP-FROB-91ee9c` control C5; Lean source commit `2143dda2b698cf6e3fb4bafbe853af7262e3ffba`. These are mathematical statements about canonical degree-five finite fields. They do not replay a scientific trial or establish an ECDLP performance claim.

Base field	Nontrivial factors of $X^5 - 1$	Lean kernel result	Sum dimension
$\mathbf{F}_{11}$	$X + 2, X + 6, X + 7, X + 8$	four successive disjoint summands = $\ker \mathrm{Tr}$	4
$\mathbf{F}_{13}$	$X^4 + X^3 + X^2 + X + 1$	sole factor kernel = $\ker \mathrm{Tr}$	4
$\mathbf{F}_{19}$	$X^2 + 5X + 1, X^2 + 15X + 1$	disjoint sum = $\ker \mathrm{Tr}$	4

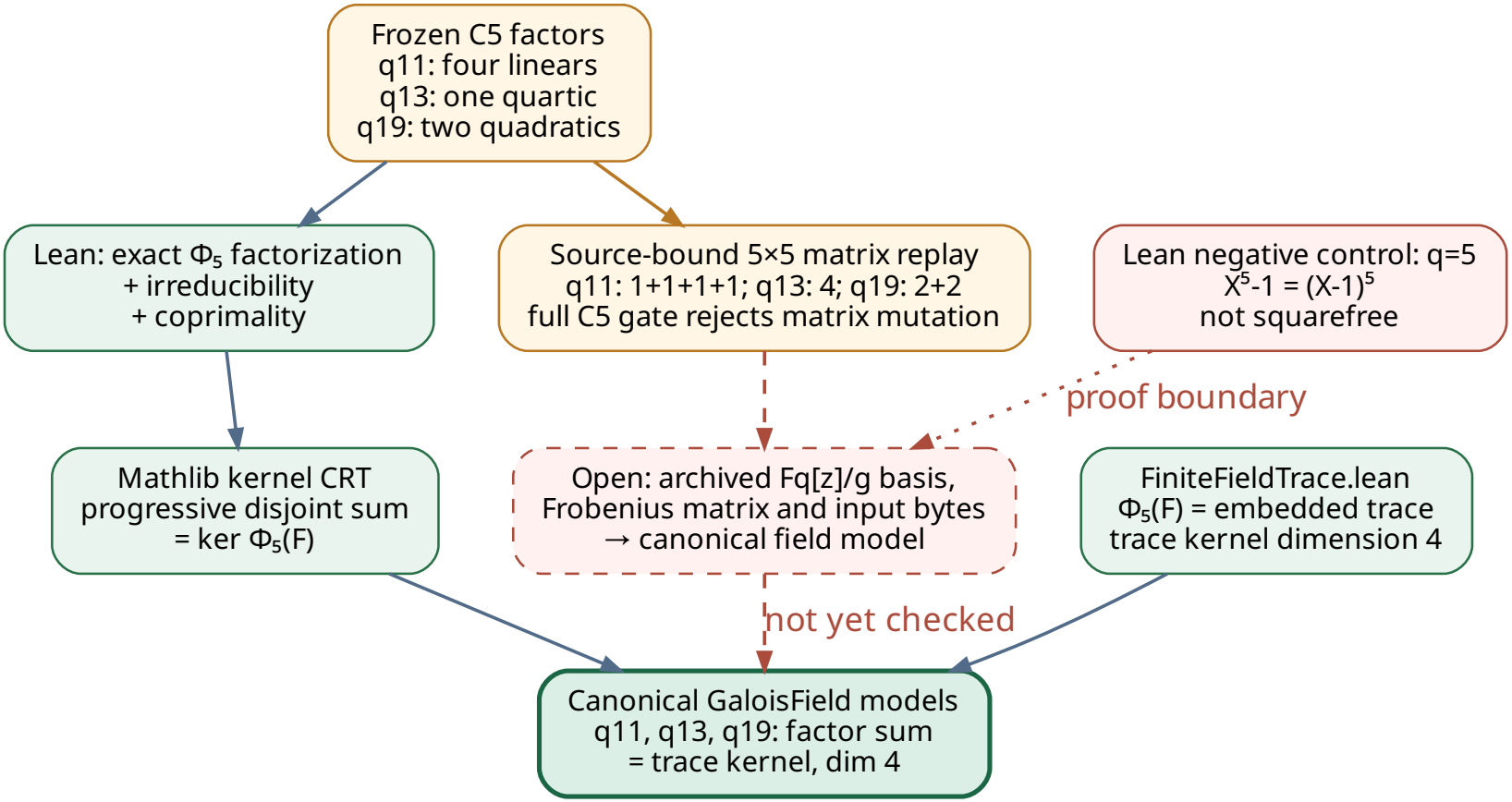
Proof method. Lean checks exact cyclotomic factorizations and irreducibility. For $q = 13, 19$, it uses Mathlib's finite-field cyclotomic factor criterion with $\mathrm{ord}_5(q) = 4, 2$; the $q = 11$ factors have degree one. Explicit coprimality certificates and Mathlib's polynomial-kernel CRT give the direct sums. The prior finite-field theorem identifies $1 + F + F^2 + F^3 + F^4$ with the embedded trace and proves its kernel has dimension four. Here F is the q-power Frobenius on `GaloisField q 5`.

Verification. The three exact-source verifier receipts accompany the report. They bind the target specifications, theorem source, pinned Lean 4.35.0-rc1 toolchain and dependency manifest. Each target axiom audit passed; the full 8,971-job build passed and 612 theorem declarations used standard axioms only. Independent semantic review remains pending.

Archived matrix replay. A separate plain-Python checker reads the three frozen outputs, recomputes each 5×5 Frobenius matrix from the recorded modulus, checks modulus/factor irreducibility and factor kernels, directness and trace equality. The same gates reject a mutated matrix in all three cells; the byte-exact receipt is included.

Negative control. A separate Lean theorem proves $X^5 - 1 = (X - 1)^5$ and non-squarefreeness over $\mathbf{F}_5$. This checks the coprimality boundary; $q = 5$ was not an archived core cell.

Boundary. No checked Lean refinement yet identifies the archived $\mathbf{F}_q[z]/(g)$ basis, executable Frobenius matrix or frozen C5 input bytes with these canonical field models. The historical run control, C8 and the experiment success gate therefore remain open. No measured value or performance dashboard curve changed.


Frozen C5 factors
q11: four linears
q13: one quartic
q19: two quadratics
Lean: exact Φ5 factorization
+ irreducibility
+ coprimality
Source-bound 5×5 matrix replay
q11: 1+1+1+1; q13: 4; q19: 2+2
full C5 gate rejects matrix mutation
Lean negative control: q=5
X5-1 = (X-1)5
not squarefree
proof boundary
Mathlib kernel CRT
progressive disjoint sum
= ker Φ5(F)
Open: archived Fq[z]/g basis,
Frobenius matrix and input bytes
→ canonical field model
FiniteFieldTrace.lean
Φ5(F) = embedded trace
trace kernel dimension 4
not yet checked
Canonical GaloisField models
q11, q13, q19: factor sum
= trace kernel, dim 4